

20250408 - Reviewing my past work (Finally)

Timeline in: [Column Generation For Regression Regularization](#)

Overview

I need to review the past work for my [master thesis](#). I think I may have some "fresh" ideas from my work with Haihao, and also from the courses at MIT, which can further improve the methodology and the implementation of the CG methods.

1. Model

It is straightforward the way to model LASSO, Ridge, and Elastic Net as a Second Order Cone Program. Let $X \in \mathbb{R}^{n \times m}$, ...

$$\begin{aligned} (SOCP) \min \quad & r + \lambda_1 e'z + \lambda_2 e'u \\ s.t \quad & \|X\beta - y\|_2 \leq r \\ & -\beta_i \leq z_i, \beta_i \leq z_i, & \forall i \in [m] \\ & \|(2\beta_i, u_i - 1)\|_2 \leq u_i + 1, & \forall i \in [m] \\ & 0 \leq u_i, & \forall i \in [m] \end{aligned}$$

To be precise, the r term should be squared to have the equivalence, and also to make easier the decomposition scheme, as

$$\begin{aligned} (SOC - E. Net) \min \quad & r^2 + \lambda_1 e'z + \lambda_2 e'u \\ s.t \quad & \|X\beta - y\|_2 \leq r \\ & -\beta_i \leq z_i, \beta_i \leq z_i, & \forall i \in [m] \\ & \|(2\beta_i, u_i - 1)\|_2 \leq u_i + 1, & \forall i \in [m] \\ & 0 \leq u_i, & \forall i \in [m] \end{aligned}$$

Although, the constraints from LASSO and Ridge regression themselves are not necessarily hard-to-handle in a relaxation scheme, since the only coupling constraint is the first constraint. The rest of the constraint are: (1) very sparse/independent without coupling constraint, and (2) as tight as possible in terms of not aggregating a constraint in a single one.

1.1 Relaxation

Hence, the only objective of the problem is to relax the first SOC constraint on a relaxation step, which also allows us to get rid of the squared term r^2 of the objective function since it has **closed-form solution** with $r = \frac{\mu}{2}$.

$$\begin{aligned}
 (L) \min_{\beta, z, u} \quad & \lambda_1 e'z + \lambda_2 e'u + \psi'X\beta - \psi'y - \left(\frac{\mu}{2}\right)^2 \\
 \text{s.t.} \quad & -\beta_i \leq z_i, \beta_i \leq z_i, & \forall i \in [m] \\
 & \|(2\beta_i, u_i - 1)\|_2 \leq u_i + 1, & \forall i \in [m] \\
 & 0 \leq u_i, & \forall i \in [m]
 \end{aligned}$$

1.2 Relaxation subproblems

Furthermore, the relaxed subproblems become **completely independent/trivially separable**, with each one of them solving a subproblem of the form

$$(L_i) \min_{\beta_i} (\psi'X)_i \beta_i + \lambda_1 |\beta_i| + \lambda_2 \beta_i^2,$$

which has a closed-form solution, depending on the sign of $(\psi'X)_i$, as

- If $\lambda_1 < |(\psi'X)_i|$, then $\beta_i = \frac{-(\psi'X)_i + \text{sign}[(\psi'X)_i] \cdot \lambda_1}{2\lambda_2}$.
- If $\lambda_1 \geq |(\psi'X)_i|$, then $\beta_i = 0$.

1.2.1 LASSO and Ridge regression extension.

The condition for those two problems is exactly the same, but setting respective opposite $\lambda_i = 0$. In other words:

1. **LASSO:** We set $\lambda_2 = 0$, to get
 - If $\lambda_1 < |(\psi'X)_i|$, then $\beta_i \rightarrow -\text{sign}[(\psi'X)_i] \cdot \infty$ (**unbounded subproblem**).
 - If $\lambda_1 \geq |(\psi'X)_i|$, then $\beta_i = 0$.
2. **Ridge:** We set $\lambda_1 = 0$, to get (a **single condition**)
 - If $0 < |(\psi'X)_i|$, then $\beta_i = \frac{-(\psi'X)_i}{2\lambda_2}$.
 - If $0 = |(\psi'X)_i|$, then $\beta_i = 0$.

⚠ Note: We also need to verify if it is worth to use SOC formulations for β_i^2 terms on the master problem or not, in terms of performance.

1.2.1.1 LASSO: Unbounded subproblem case

Let us continue this section on a new notebook: [202504 - LASSO - Unbounded subproblem case](#).

Tags

#Tesis #Uchile #ColumnGeneration